Active Structural Control in Nonlinear Field Systems:

Parameter-Dependent Steering of Coherence

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Abstract

The Critical Coherence Index (CCI) has been introduced as a structural diagnostic that quantifies the balance between dynamical activity and coherence in nonlinear systems. While previous work established its scaling behaviour, predictive capability, and a passive dynamical realisation via a coherence field, the framework remained fundamentally observational: it describes which structures emerge but not how they can be actively influenced.

In this work, we extend the CCI framework into an active control paradigm. By introducing a feedback control term into the coherence-field dynamics, we show that structural coherence can be treated as a controllable quantity rather than a purely emergent observable. The complete feedback loop $\phi \to C \to U \to \phi$ is formalised, and a controlled evolution equation is derived that enables target-driven steering of the system in coherence space.

Numerical experiments in two-dimensional ϕ^4 field systems demonstrate that active control can simultaneously reduce structural free energy, enhance nearest-neighbour mutual information, and induce measurable changes in the CCI. A systematic parameter sweep reveals a nontrivial control window, in which the effect is strongest in the low-damping regime. In this regime, control transitions from structural refinement to genuine regime steering.

These results indicate that structural selection can be extended from a passive diagnostic principle to an active control law.

1 Introduction

Nonlinear field systems, including scalar ϕ^4 models [21, 9], exhibit a wide range of dynamical behaviours, including ordered structures, critical regimes, and chaotic dynamics. Recent work introduced the Critical Coherence Index (CCI) as a structural observable quantifying the balance between dynamical activity and coherence [3, 1].

A minimal dynamical realisation of coherence was introduced in a companion paper [2] via a coarse-grained field $C(x,t)$ obeying:

$$\partial_t C = D\nabla^2 C + aC - bC^3. \tag{1}$$

This formulation captures structural selection as a gradient-flow process. However, Eq. (1) is *passive*: it describes which structures emerge, but not how to actively steer a system toward desired structural states. The present paper extends this passive framework into an active control paradigm, constituting a direct continuation of [2]. Feedback control of pattern-forming systems has been studied extensively in related frameworks, including the complex Ginzburg–Landau equation [6, 7, 8]; the present work applies analogous ideas to the ϕ^4 field system using coherence as the controlled quantity.

Core question: Can structural coherence be controlled rather than merely observed?

2 Feedback Loop and Coupling Structure

A central feature of the proposed control scheme is the closed feedback loop between the ϕ-field and the coherence field C. The coupling is as follows:

1. **Field to coherence:** $C(x,t)$ is computed from $\phi(x,t)$ as a spatially coarse-grained measure of local dynamical activity, specifically via the local variance of ϕ over a coarse-graining window:
$$C(x,t) = \mathcal{G}_\sigma[\phi^2](x,t) - \big(\mathcal{G}_\sigma[\phi](x,t)\big)^2, \tag{2}$$
Here, $\mathcal{G}_\sigma$ denotes a Gaussian smoothing kernel of width σ. The resulting field $C(x,t)$ can thus be interpreted as a local measure of structural fluctuations or dynamical activity.

   Note that the field $C(x,t)$ defined in Eq. (2) serves as the operational control variable in the feedback loop, whereas the CCI is evaluated separately as a diagnostic observable. These quantities are related but distinct: $C(x,t)$ captures local coarse-grained variance, while the CCI tracks reduced structural instability at the level of diagnostic observables. A canonical clarification of this distinction is given in Ref. [5].

2. **Coherence to control:** The control signal is computed as
$$U(x,t) = -(C(x,t) - C^*), \tag{3}$$
where C^* is a prescribed target coherence level.

3. **Control to field:** The control signal acts additively on the ϕ-field dynamics:
$$\ddot{\phi} = \nabla^2\phi - (\phi^3 - \phi) - \gamma\dot{\phi} + \lambda U(x,t). \tag{4}$$

The complete feedback loop is therefore:

$$\phi \xrightarrow{\text{Eq. (2)}} C \xrightarrow{\text{Eq. (3)}} U \xrightarrow{\text{Eq. (4)}} \phi. \tag{5}$$

This loop is evaluated at every time step of the numerical integration, ensuring continuous feedback between structural state and control action.

3 Controlled Coherence Field Dynamics

Extending Eq. (1) with the control term yields:

$$\partial_t C = D\nabla^2 C + aC - bC^3 + \lambda U(x,t), \tag{6}$$

where $U(x,t)$ is the feedback signal defined in Eq. (3). This can be interpreted as a linear restoring force in coherence space, driving C toward the target C^*. The controlled C-equation should be understood as an effective coarse-grained description consistent with, but not derived exactly from, the underlying controlled ϕ-dynamics (Eq. (4)). A first-principles derivation of this control term from the ϕ-dynamics is subject to ongoing work.

4 Methods

We simulate a two-dimensional ϕ^4 field on a periodic lattice using the controlled dynamics defined above, with explicit time integration.

4.1 Numerical Scheme

The field evolution is discretised using finite differences and integrated in time using a fourth-order Runge–Kutta (RK4) scheme.

4.2 Control Algorithm

At each time step, the system evolves according to the feedback loop formalised in Section 2:

1. Compute $C(x,t)$ from $\phi(x,t)$ via Eq. (2).
2. Evaluate structural diagnostics (CCI, F_{struct}, I_{nn}).
3. Compute control signal $U(x,t)$ via Eq. (3).
4. Update field using RK4 integration of Eq. (4).
5. Repeat for all time steps.

4.3 Target Coherence Level

The target C^* is set to a spatially uniform constant equal to the mean coherence of the ordered-regime reference configurations identified in [1]. This choice is motivated by the observation that ordered configurations consistently exhibit low CCI values and low F_{struct}; setting C^* to this reference level therefore steers the system toward the ordered regime without requiring a dynamically evolving target. The sensitivity to alternative choices of C^* is an open question left for future work.

4.4 Composite Score

To compare parameter settings across the sweep, we define a composite score combining relative improvements across the three primary observables:

$$\mathcal{S}(\gamma,\lambda) = w_1 \frac{\Delta\text{CCI}}{\text{CCI}_0} + w_2 \frac{\Delta F_{\text{struct}}}{|F_{\text{struct},0}|} + w_3 \frac{\Delta I_{\text{nn}}}{I_{\text{nn},0}}, \tag{7}$$

where $\Delta(\cdot) = (\cdot)_{\text{controlled}} - (\cdot)_{\text{baseline}}$ and the weights are $w_1 = 0.4$, $w_2 = 0.4$, $w_3 = 0.2$. A lower value of $\mathcal{S}$ indicates better overall structural improvement (note that improvements correspond to $\Delta\text{CCI} < 0$ and $\Delta F_{\text{struct}} < 0$, so lower $\mathcal{S}$ is preferable).

4.5 Simulation Parameters

Table 1 summarises all numerical parameters used in the simulations, as specified in the sweep configuration.

Table 1: Numerical parameters for all Paper 19 simulations.

Parameter	Description	Value
$N_x \times N_y$	Lattice size	64×64
Δx	Spatial step	1.0
Δt	Time step (RK4)	0.02
N_{steps}	Integration steps	1800
T_{total}	Total simulation time	36.0
σ_C	Gaussian coarse-graining width	1 (lattice unit)
D	Effective coherence diffusion coefficient	1 (dimensionless scaled units)
a	Linear growth coefficient	1.0 (scaled)
b	Cubic saturation coefficient	1.0 (scaled)
w_{wall}	Domain-wall target width	6.0
γ	Damping (sweep)	[0.02, 0.03, 0.04, 0.06, 0.08]
λ	Control strength (sweep)	[0.0, 0.08, 0.16, 0.24, 0.32]
seed	Random seed (fixed)	42

4.6 Parameter Sweep

A grid search is performed over:

$$\gamma \in [0.02, 0.08], \quad \lambda \in [0.0, 0.32].$$

Each configuration is compared to an uncontrolled baseline using identical initial conditions.

5 Results

We performed a systematic parameter sweep over damping γ and control strength λ in the controlled two-dimensional ϕ^4 benchmark. For each damping value, an uncontrolled baseline run was compared against controlled runs with increasing control strength.

The sweep reveals a clear and nontrivial dependence on both parameters. Active structural control is most effective in the low-damping regime ($\gamma \lesssim 0.03$), whereas at higher damping its effect is primarily limited to structural refinement without significant regime shifts.

The heatmaps shown in Figs. 1–3 confirm that the strongest control response occurs in the low-damping regime. The best-performing configuration is $\gamma = 0.02$, $\lambda = 0.24$, identified via the composite score $\mathcal{S}$ defined in Eq. (7). In this case:

- CCI decreases from 0.0579 to 0.0540,
- F_{struct} decreases from 3.3433 to 3.1322,
- I_{nn} increases from 0.1080 to 0.2746.

Across the full sweep, F_{struct} decreases for nearly all nonzero λ, while I_{nn} increases systematically. The response of the CCI is more selective, with clear reductions occurring primarily at low damping.

This behaviour suggests two operational regimes:

- *Structure refinement* ($\gamma \gtrsim 0.05$): control improves F_{struct} and I_{nn} without significant CCI shifts,
- *Regime steering* ($\gamma \lesssim 0.03$): control induces measurable CCI reduction, indicating a genuine structural regime transition. These structural transitions are reminiscent of kink and domain-wall dynamics in nonlinear field systems [10, 20].

A quantitative boundary between these regimes is not yet determined; we estimate the crossover at $\gamma_c \approx 0.03$–0.04 based on the observed qualitative change in CCI response, but a rigorous determination of γ_c requires further systematic analysis.

Robustness. Selected parameter configurations ($\gamma = 0.02$, $\lambda = 0.24$ and $\gamma = 0.06$, $\lambda = 0.16$) were evaluated across $n = 5$ independent random initial conditions. The observed trends in F_{struct} and I_{nn} remain qualitatively consistent across runs, with minor quantitative variations. Given the small sample size, these results should be interpreted as indicative rather than statistically conclusive; a full robustness analysis with $n \geq 20$ is deferred to future work.

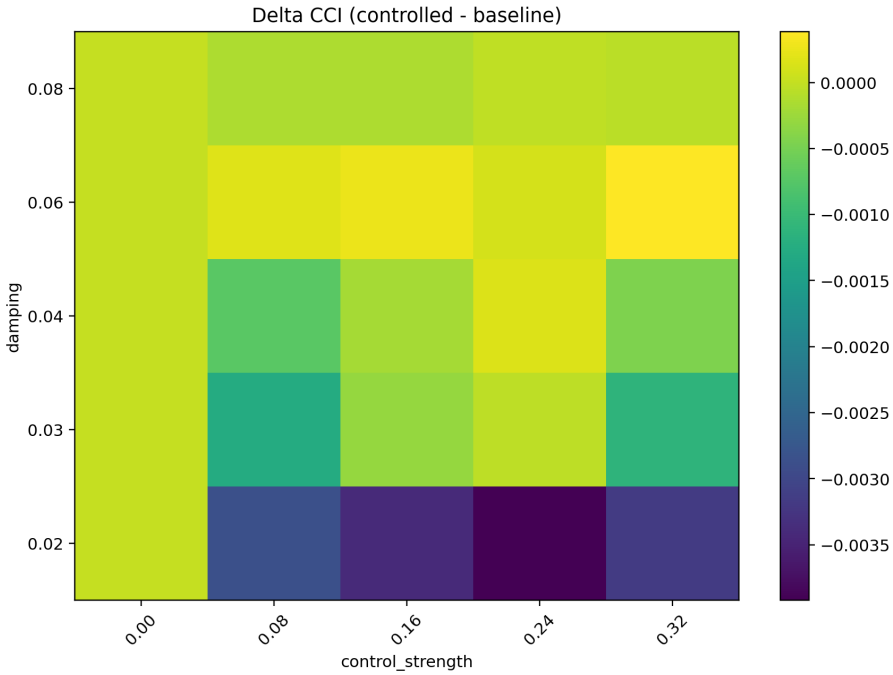

Figure 1: Heatmap of ΔCCI across damping γ and control strength λ. Negative values indicate a reduction in structural stress under active control. The strongest effect is observed in the low-damping regime ($\gamma \lesssim 0.03$).

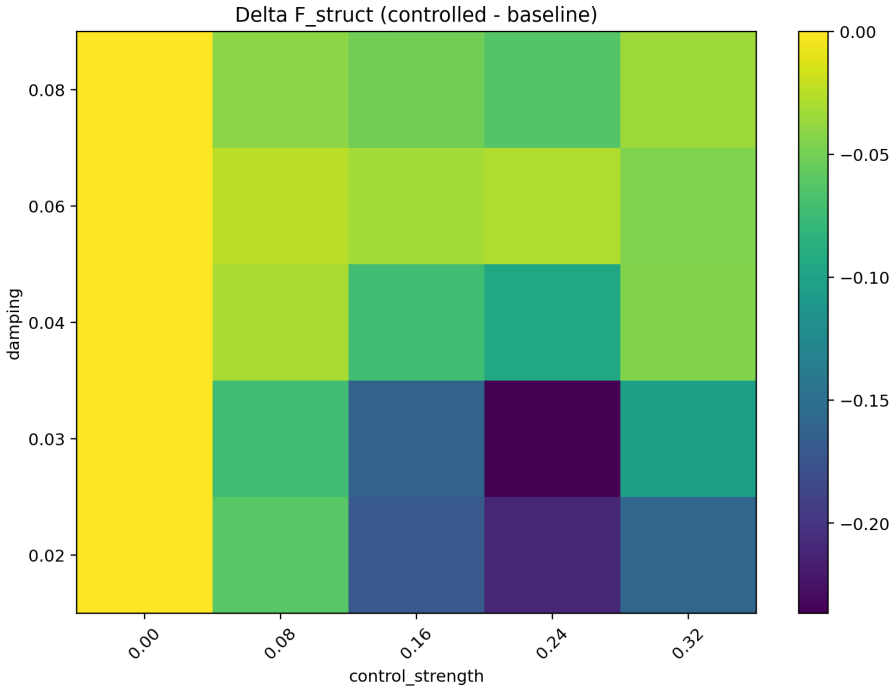

Figure 2: Heatmap of ΔF_{struct}. Negative values indicate improved structural preference. The improvement is robust across nearly all nonzero control strengths.

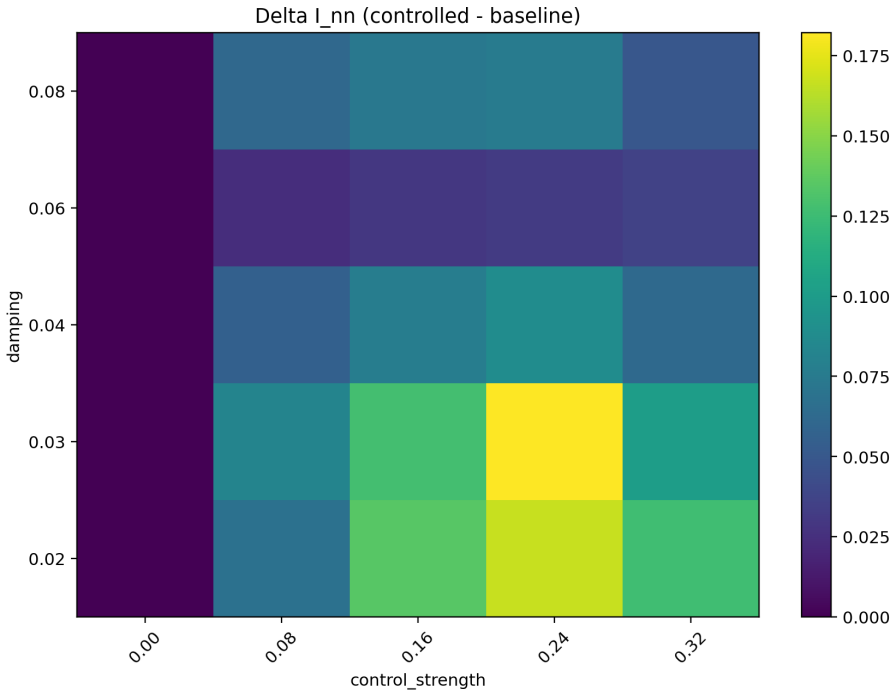

Figure 3: Heatmap of ΔI_{nn}. Positive values indicate enhanced nearest-neighbour coherence. The increase is systematic across the parameter space.

6 Comparison to Other Control Approaches

The proposed control mechanism differs conceptually from classical optimal control and model predictive control (MPC), which typically rely on explicit cost function minimisation over tra-

jectories [13]. In contrast, the present approach operates directly in coherence space via a local feedback law, without requiring global trajectory optimisation.

Compared to pinning control methods, which fix specific degrees of freedom, the proposed method acts globally through a distributed coherence field, enabling smoother structural transitions [12].

For a discussion of general observability and controllability in nonlinear networks see [17]; for nonlinear control in related mechanical systems see [18]. We note that this section provides a *conceptual positioning* rather than a quantitative comparison; a direct numerical benchmarking against MPC or pinning control is beyond the scope of the present paper and is identified as a direction for future work.

7 Physical Realisability

The control term can be interpreted as an externally applied feedback mechanism acting on the system's coherence structure. Possible physical realisations include:

- *Nonlinear optical systems*: coherence can be manipulated via external fields and coherence synthesis [14, 19],
- *Josephson junction arrays*: phase coherence is externally tunable through applied currents and coupling [11],
- *Soft-matter systems*: effective interaction potentials can be controlled via optical tweezers or external field gradients.

In all cases, the control parameter λ corresponds to the strength of external driving or coupling to a feedback loop. We emphasise that these physical analogies are suggestive rather than rigorous, and that a direct experimental implementation of the proposed scheme has not yet been realised.

8 Interpretation

The central insight of this work is:

> Structural coherence is not only an observable, but a controllable quantity. The numerical results demonstrate that this control is both measurable and parameter-dependent, with qualitatively distinct regimes of effectiveness.

9 Implications

The control formulation suggests potential applications in adaptive control of complex systems, including machine learning frameworks where structural coherence plays a role analogous to regularisation. A connection to the Free Energy Principle [15, 16] may also be explored, as variational free-energy minimisation shares structural features with the proposed feedback mechanism.

10 Reproducibility

All simulations and analysis routines are available at:

`https://github.com/Chris4081/structural-selection-principle`

Simulations were performed with a fixed random seed and default parameter settings as specified in the repository. The provided scripts reproduce all figures and data tables reported in this work.

11 Limitations

The present framework has several important limitations:

- The feedback loop is phenomenological; a first-principles derivation connecting the ϕ-field dynamics to the coherence field via a rigorous coarse-graining procedure is not yet established.
- The target C^* is set to a spatially uniform constant; the sensitivity to non-uniform or dynamically evolving targets has not been explored.
- The composite score $\mathcal{S}$ involves heuristic weights; alternative weightings may shift the identified optimum.
- Robustness has been assessed for $n = 5$ initial conditions only; full statistical validation requires larger ensembles.
- The crossover damping γ_c between refinement and steering regimes is estimated qualitatively; a rigorous determination is open.

12 Conclusion

We have extended the CCI framework from a passive structural diagnostic into an active control paradigm. By formalising the complete feedback loop $\phi \to C \to U \to \phi$ and introducing an explicit composite score for parameter comparison, we provide a methodologically transparent basis for active structural control in nonlinear field systems.

Numerical experiments confirm that the control scheme can steer structural observables in a parameter-dependent manner, with two qualitatively distinct operational regimes: structural refinement at high damping and regime steering at low damping.

These results constitute a first step toward actively steering structure formation in nonlinear field systems, with potential extensions to complex adaptive systems and AI-relevant structural optimisation.

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A Relation Between the Control Field C and the CCI

The present paper does not redefine the Critical Coherence Index. Instead, it uses two related but distinct levels of description.

First, the field $C(x,t)$ in Eq. (2) is the *operational control variable*. It is constructed as a Gaussian-smoothed local variance of ϕ and therefore provides a spatially resolved quantity that can be fed directly into the feedback law.

Second, the CCI is used as a *diagnostic observable* evaluated alongside F_{struct} and I_{nn} during the controlled evolution. In the canonical clarification developed later in Ref. [5], the CCI is organised on three distinct layers:

$$\text{formal definition} \longrightarrow \text{physical proxy} \longrightarrow \text{empirical scaling.}$$

Within that hierarchy, the present control paper should be read at the level of an operational benchmark: $C(x,t)$ drives the feedback loop, while the CCI is monitored as a reduced structural instability indicator. The two quantities are physically related, but they are not identical and need not share the same operational formula.